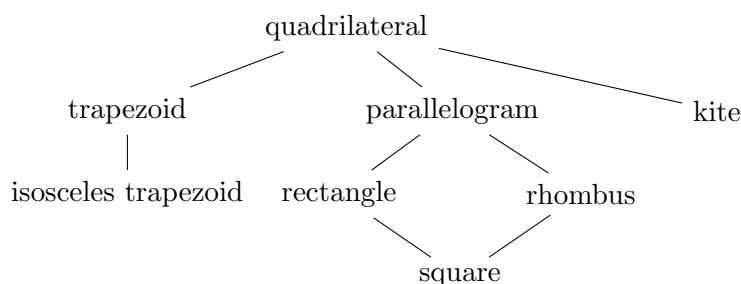


Classifying Quadrilaterals from Their Properties

Directions

- The interior angles of any quadrilateral add to 360° .
- **Most specific name** means the name furthest down the family tree below that the stated properties actually force. If a property is not given, you may not assume it.
- Every classification answer must be *justified*: name the property you used and say what it rules in or out.
- In this booklet a *trapezoid* has exactly one pair of parallel sides.



The quadrilateral family. A name lower down carries every property of the names above it. No measurements are shown — use the properties given in each problem.

1. A quadrilateral has exactly one pair of parallel sides, and its four interior angles measure $(3x)^\circ$, $(3x)^\circ$, $(2x)^\circ$ and $(2x)^\circ$.

(a) Find x .

Answer: _____

(b) Give the most specific name for this quadrilateral, and justify it from the properties you were given.

2. The diagonals of quadrilateral $JKLM$ cross at a point that divides each of them into two equal parts. On one diagonal those two parts measure $(2x + 1)$ cm and 13 cm.

(a) Find x .

Answer: _____

(b) Give the most specific name that the stated property forces, and justify it.

3. A quadrilateral has four congruent sides. One side measures $(3x - 4)$ cm and another measures 20 cm. Its two diagonals are also congruent to each other.

(a) Find x .

Answer: _____

(b) Give the most specific name for this quadrilateral, and justify it from both stated properties.

4. In quadrilateral $ABCD$ both pairs of opposite angles are congruent. The four interior angles measure $(2x + 10)^\circ$, $(3x - 20)^\circ$, $(2x + 10)^\circ$ and $(3x - 20)^\circ$.

(a) Find x .

Answer: _____

(b) Find the measure of the larger pair of angles, in degrees.

Answer: _____

(c) Classify $ABCD$ as specifically as the information allows, and state one extra piece of information that would make it a rectangle.

5. Decide whether each statement is *always*, *sometimes* or *never* true, and explain your decision from the defining properties.

(a) A rhombus is a square.

(b) A square is a rhombus.

(c) A trapezoid is a parallelogram.

6. The diagonals of quadrilateral $EFGH$ are congruent to each other and bisect each other, but they are *not* perpendicular. One diagonal measures $(4x + 5)$ cm and the other measures 33 cm.

(a) Give the most specific name for $EFGH$, and justify it.

(b) Find x .

Answer: _____

(c) Explain why $EFGH$ cannot be a square.

7. In quadrilateral $ABCD$, $AB = AD = (3x - 1)$ cm and $CB = CD = 14$ cm. It is also known that $AB = 17$ cm.

(a) Find x .

Answer: _____

(b) Give the most specific name for $ABCD$, and explain why it is not a rhombus.

8. A student claims that every quadrilateral with two pairs of congruent sides must be a parallelogram.

(a) Give a counterexample: name a quadrilateral that has two pairs of congruent sides but is not a parallelogram, and describe how its congruent sides are arranged.

(b) In a different quadrilateral one pair of opposite sides measures $(5x - 8)$ cm and 27 cm. Find the value of x that makes those two sides congruent.

Answer: _____

(c) The other pair of opposite sides measures 19 cm and $(2y + 3)$ cm. Find the value of y that makes those two sides congruent.

Answer: _____